Lecture 1:
Introduction to Machine Learning

ML/AI for Engineering Applications

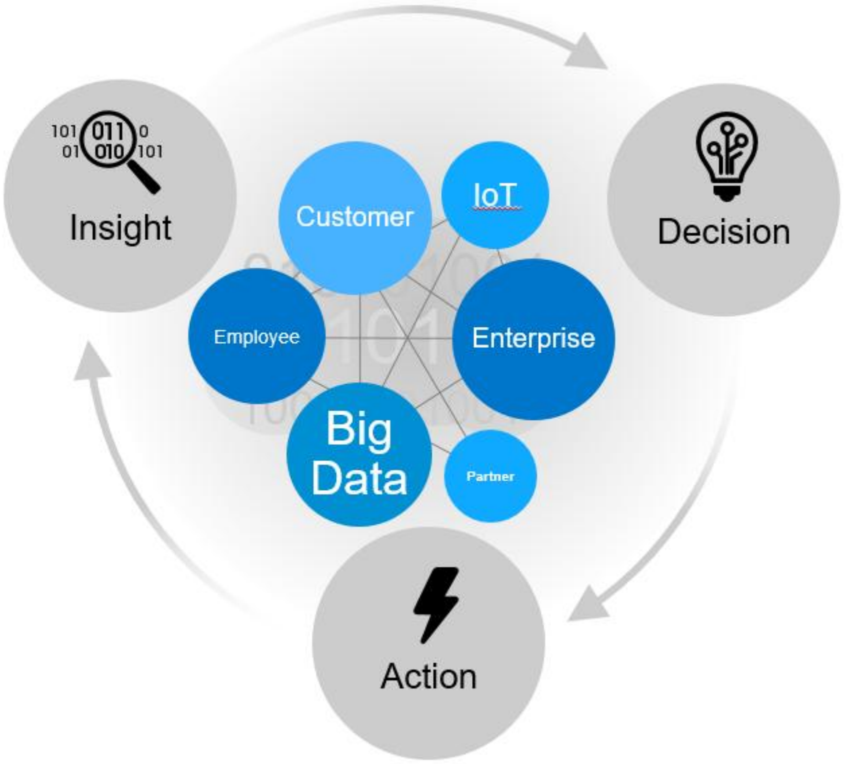
ENV 6932

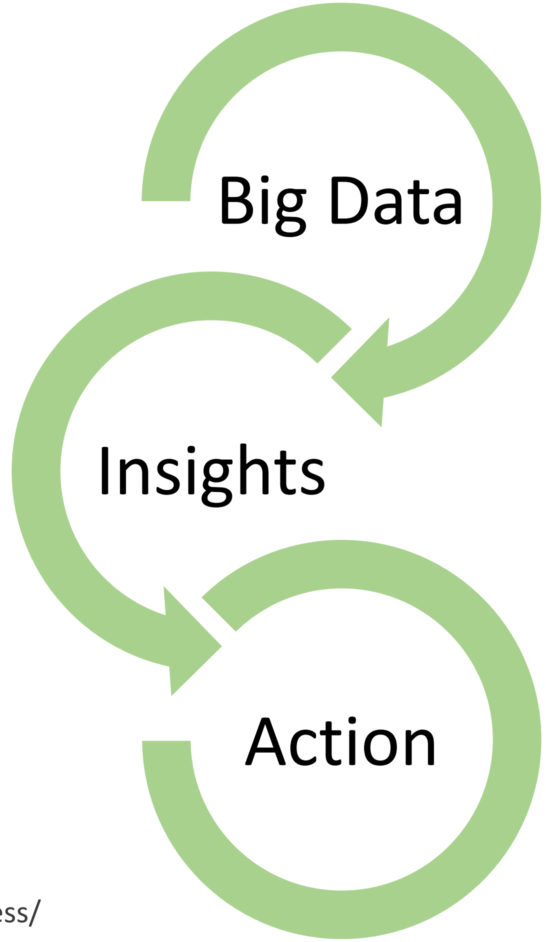
Sara Behdad

Fall 2025

Why Data Science?

Data + Analytics + Action= Business Success

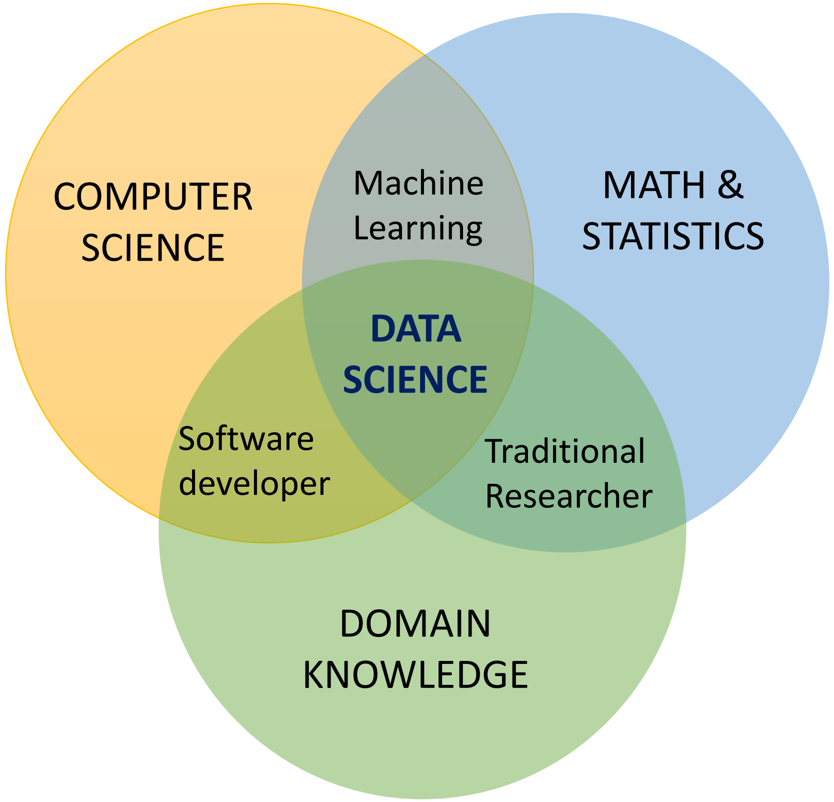




https://itelligencegroup.com/cn/global-blog/combining-the-power-of-data-and-analytics-to-drive-business-success/

What is Data Science?

- Combination of three expertise: business knowledge, statistical analysis, and computer science.



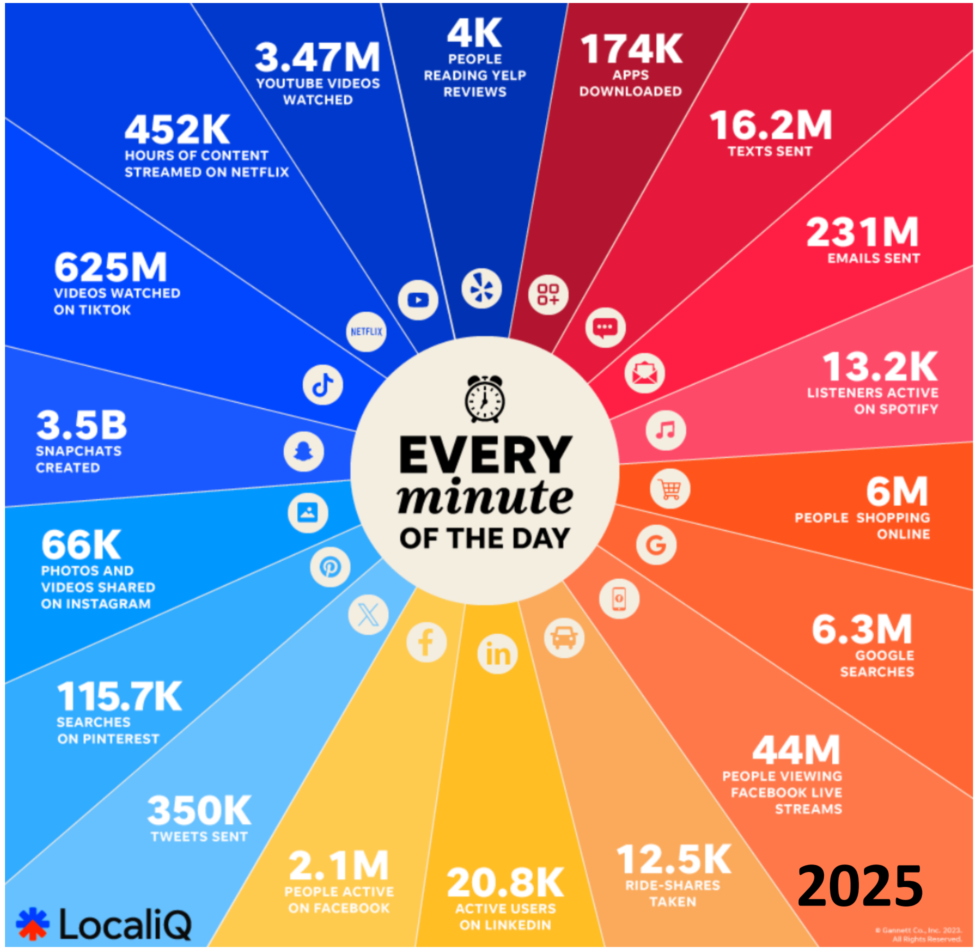
https://www.edureka.co/blog/what-is-data-science/

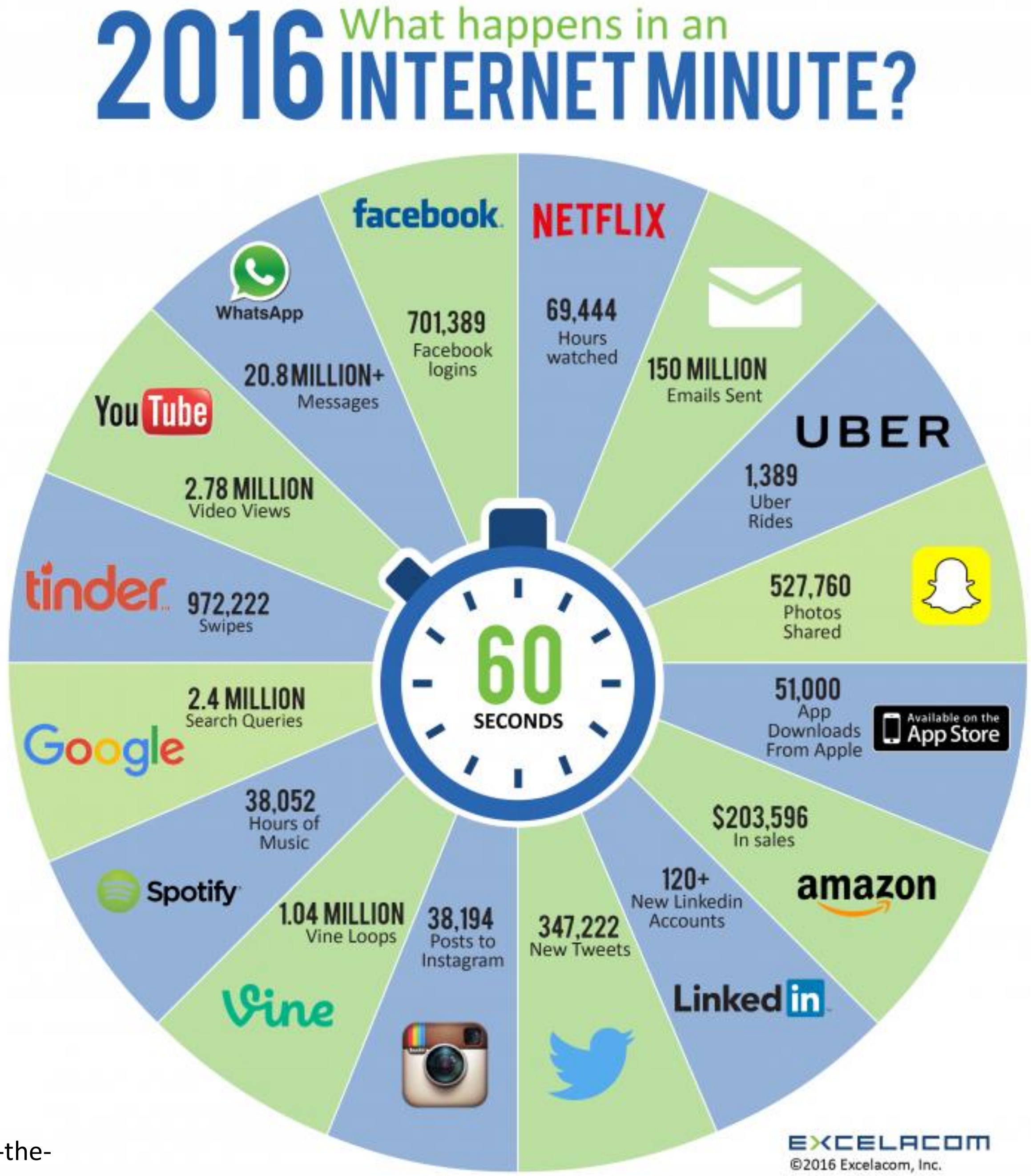
One of the most demanding job of the 21st Century?

- According to Harvard Business Review:
- **Data scientists** are difficult and expensive to hire and, given the very competitive market for their services, difficult to retain.
- There simply aren't a lot of people with their combination of scientific background and computational and analytical skills.

https://hbr.org/2012/10/data-scientist-the-sexiest-job-of-the-21st-century
by Thomas H. Davenport and D.J. Patil

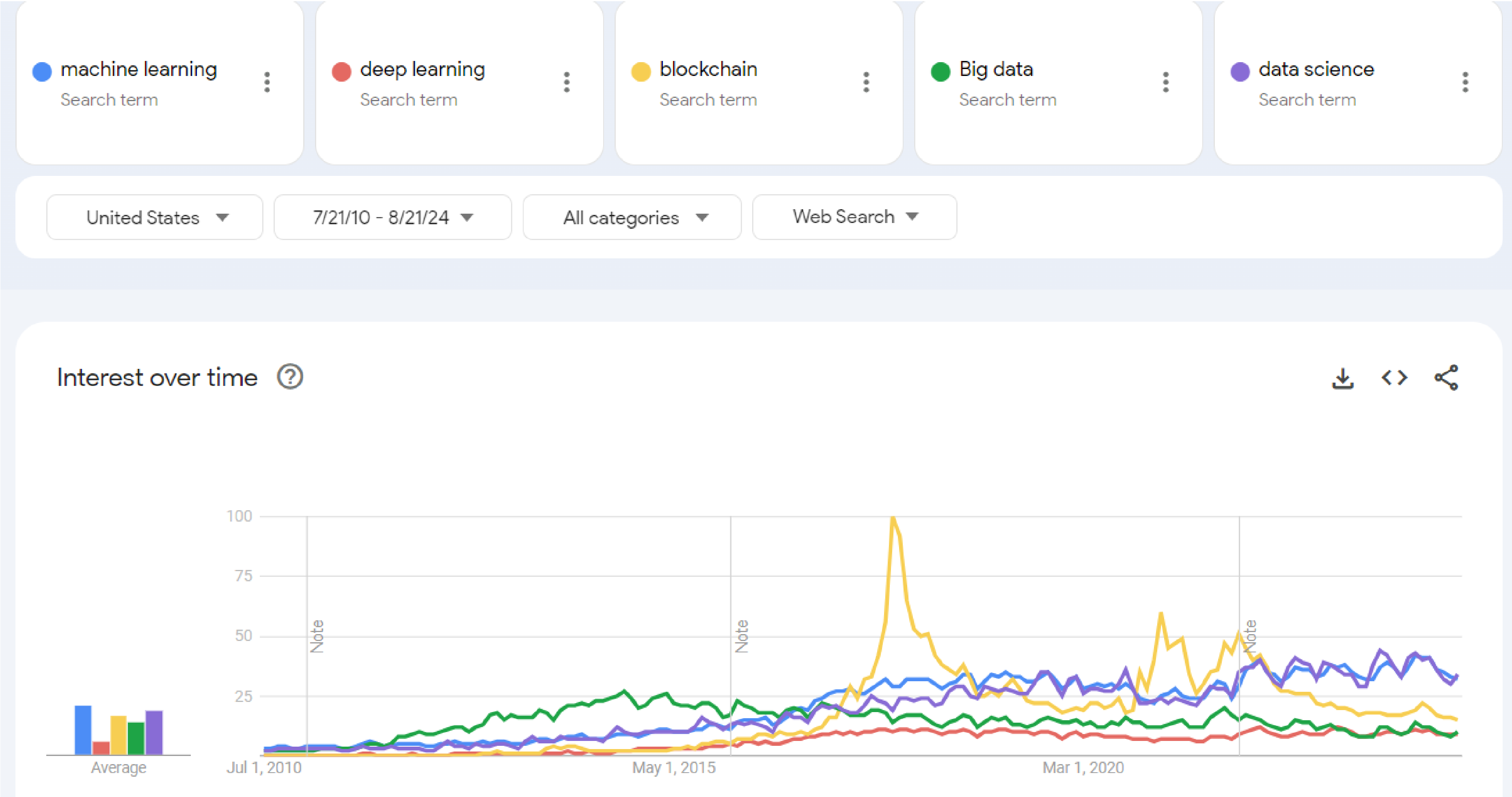
This is what happens on the Internet in 60 seconds





https://www.marketwatch.com/story/one-chart-shows-everything-that-happens-on-the-internet-in-just-one-minute-2016-04-26

The Growth of Interest in Data Science



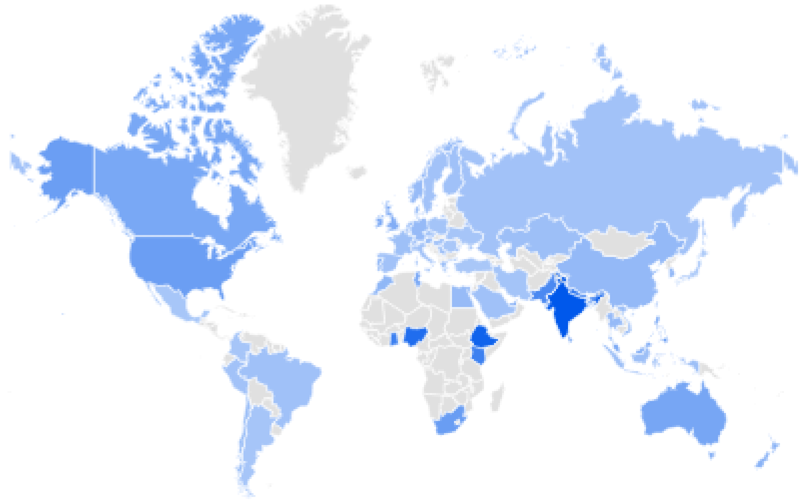
Google Trends

Interest by Region

data science

Interest by region

Region



Include low search volume regions

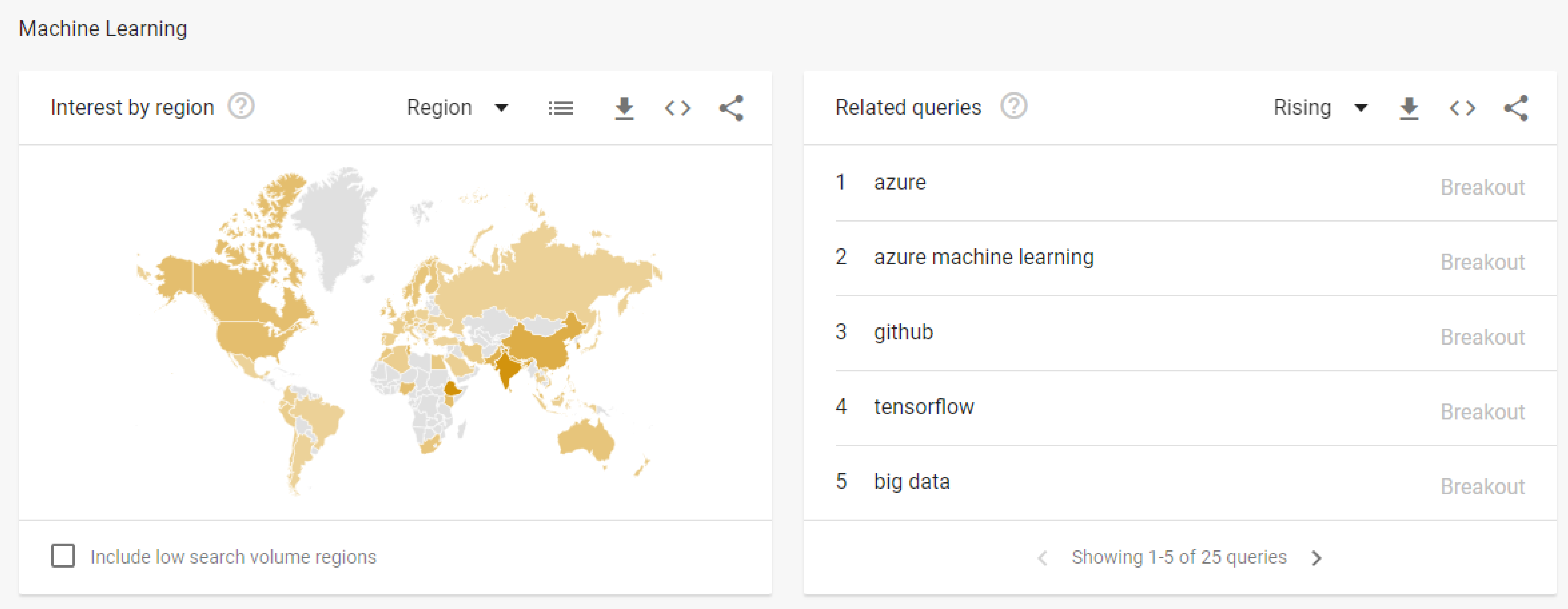
Related queries

Rising

1	coursera data science	Breakout
2	coursera	Breakout
3	data science and machine learning	Breakout
4	data science internship	Breakout
5	reddit data science	Breakout

Showing 1-5 of 25 queries

Interest by Region, Machine Learning

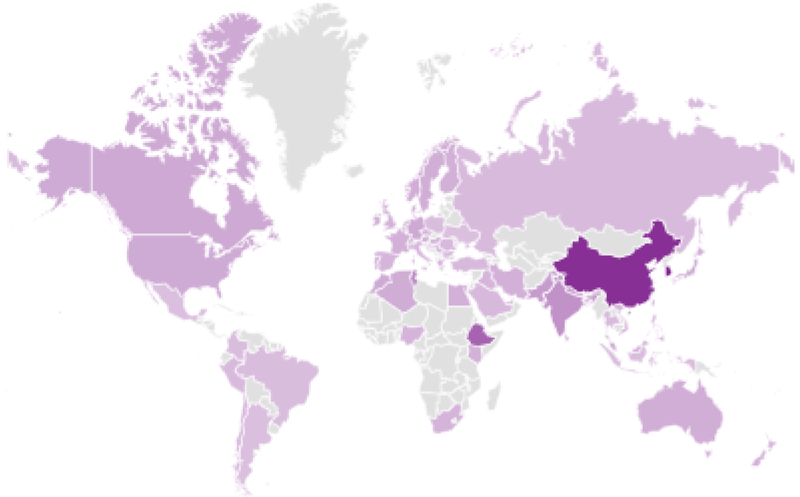


Interest by Region, Deep Learning

Deep Learning

Interest by region

Region



Include low search volume regions

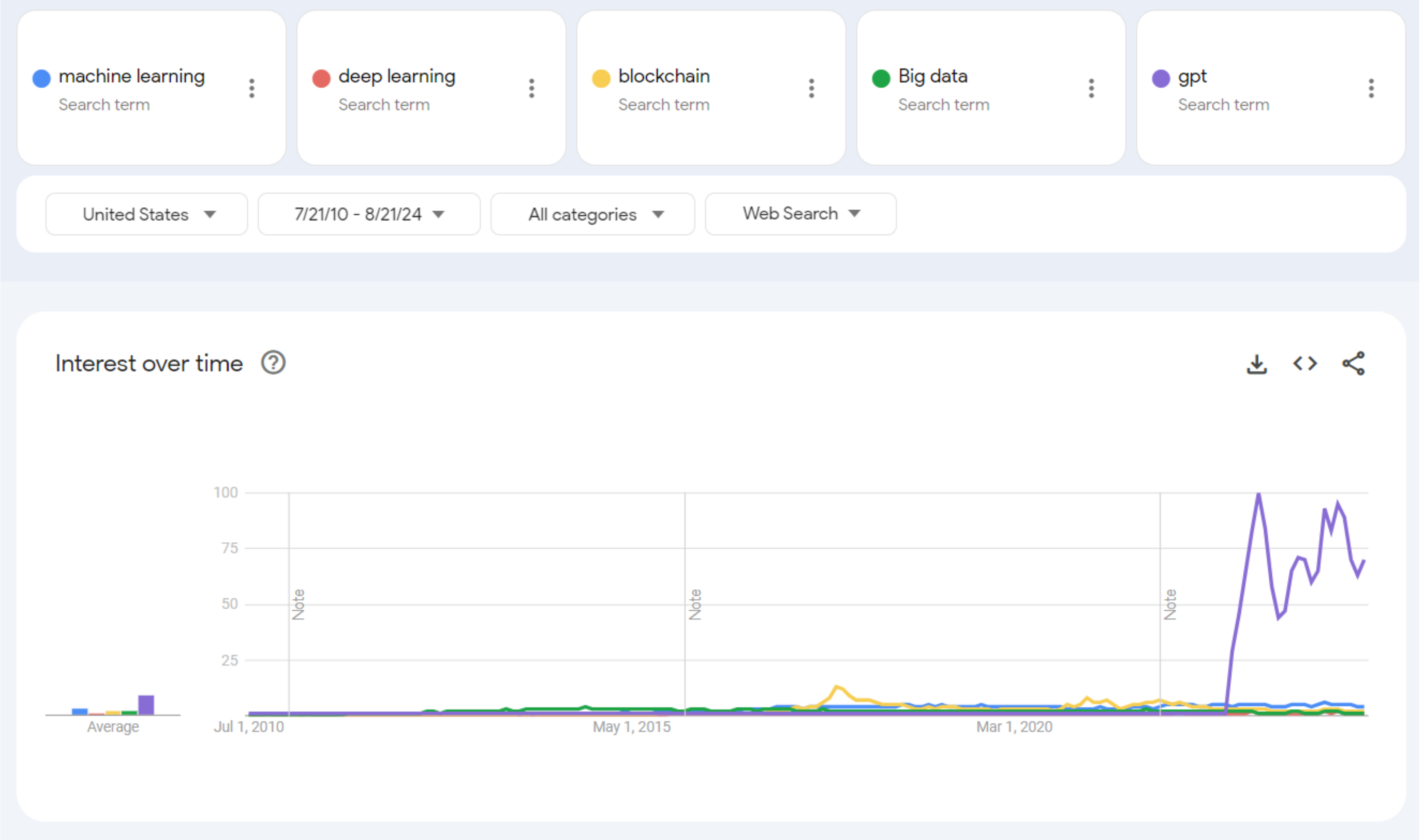
Related queries

Rising

1	neural network	Breakout
2	github	Breakout
3	deep learning tensorflow	Breakout
4	tensorflow	Breakout
5	gpu	Breakout

Showing 1-5 of 25 queries

How About GPT?

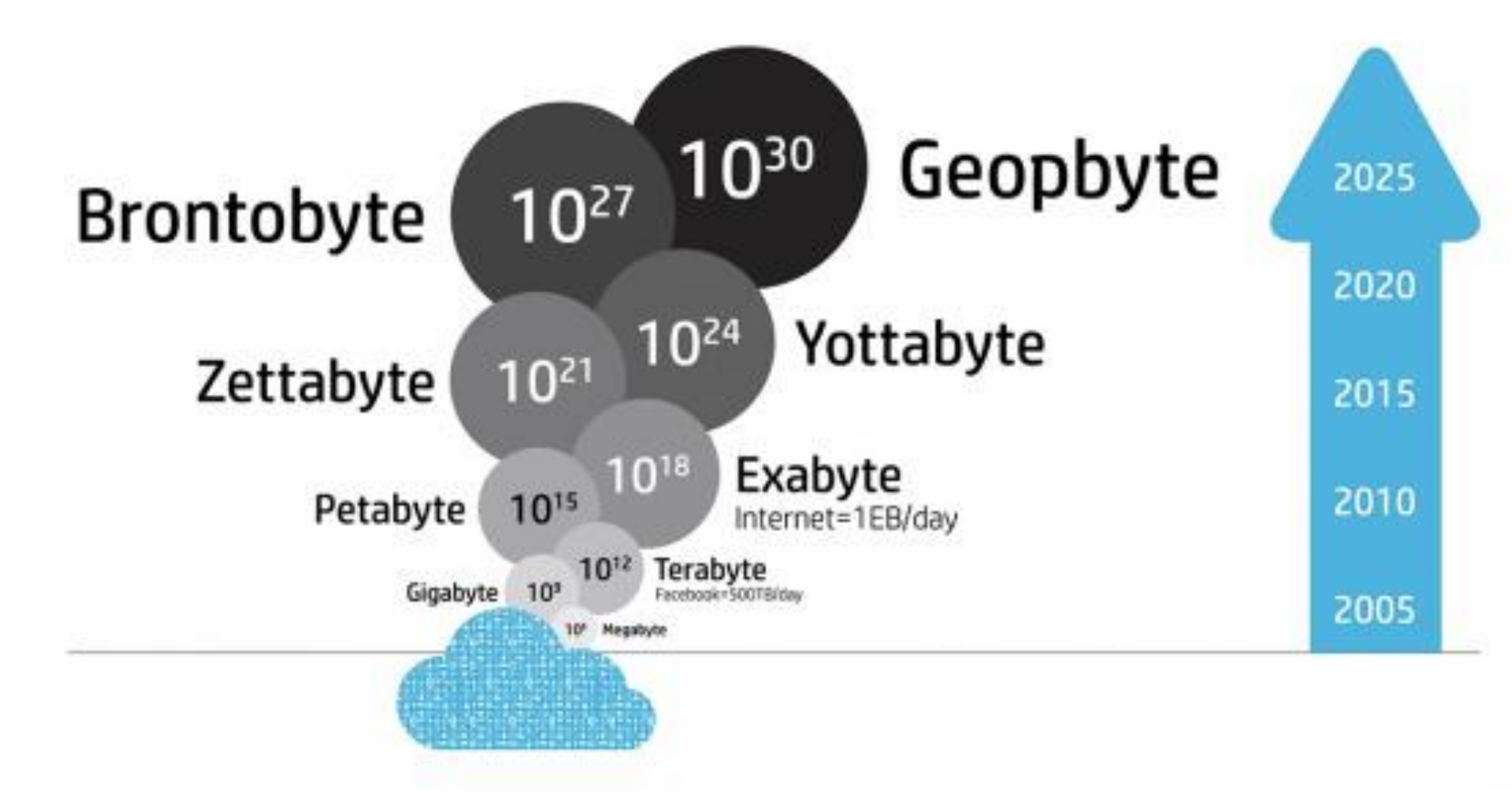


Data Volume Units

- Kilo- means 1,000; a Kilobyte is one thousand bytes.
- Mega- means 1,000,000; a Megabyte is a million bytes.
- Giga- means 1,000,000,000; a Gigabyte is a billion bytes.
- Tera- means 1,000,000,000,000; a Terabyte is a trillion bytes.
- Peta- means 1,000,000,000,000,000; a Petabyte is 1,000 Terabytes.
- Exa- means 1,000,000,000,000,000,000; an Exabyte is 1,000 Petabytes.
- Zetta- means 1,000,000,000,000,000,000,000; a Zettabyte is 1,000 Exabytes.
- Yotta- means 1,000,000,000,000,000,000,000,000; a Yottabyte is 1,000 Zettabytes.

https://mynasadata.larc.nasa.gov/basic-page/data-volume-units

How Much Data Do We Need to Deal With?



https://www.fastcompany.com/3033549/hps-post-electronic-solution-to-tomorrows-huge-data

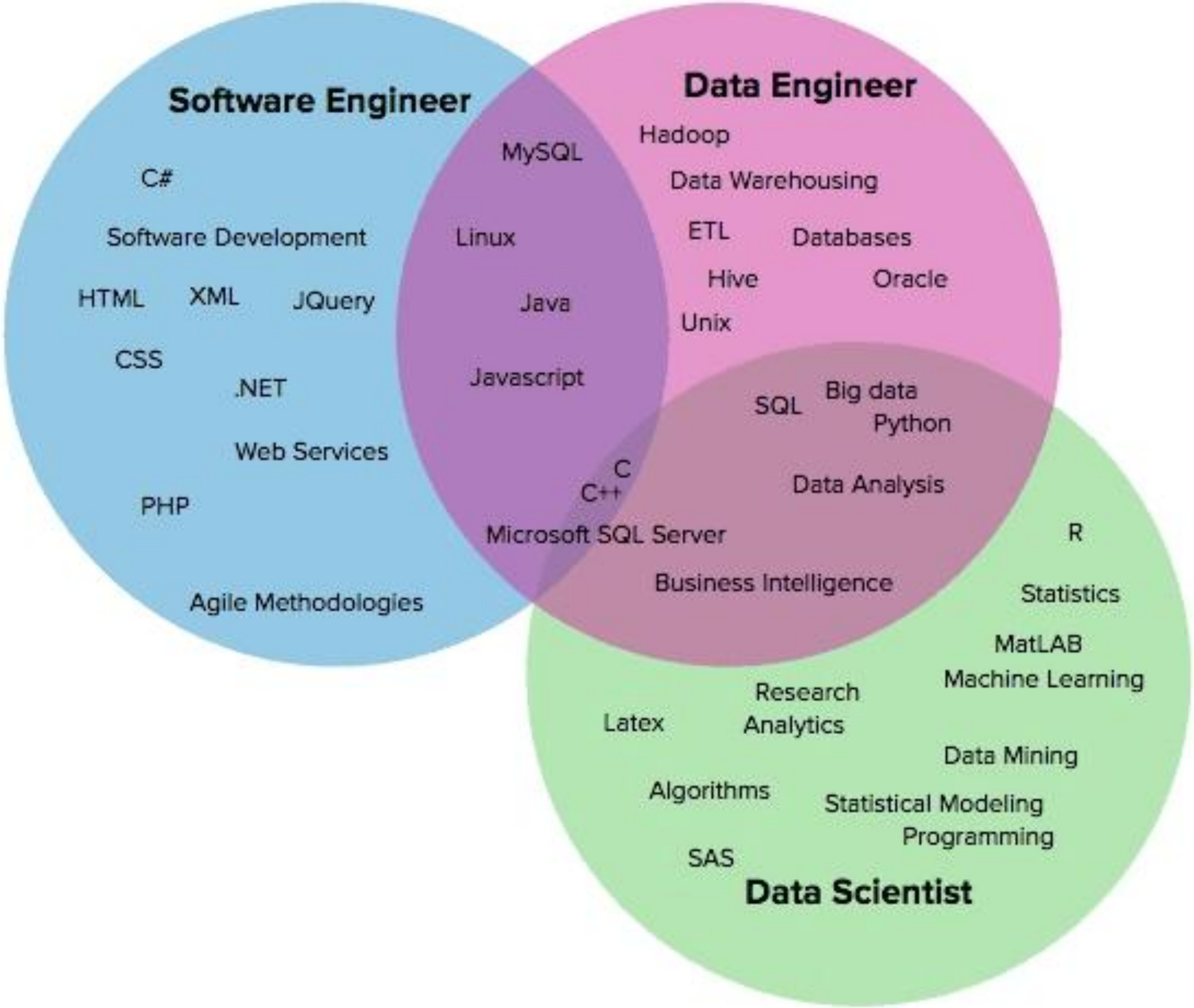
Examples of Data Volumes

Unit	Value	Example
Kilobytes (KB)	1,000 bytes	a paragraph of a text document
Megabytes (MB)	1,000 Kilobytes	a small novel
Gigabytes (GB)	1,000 Megabytes	Beethoven's 5th Symphony
Terabytes (TB)	1,000 Gigabytes	all the X-rays in a large hospital
Petabytes (PB)	1,000 Terabytes	half the contents of all US academic research libraries
Exabytes (EB)	1,000 Petabytes	about one fifth of the words people have ever spoken
Zettabytes (ZB)	1,000 Exabytes	as much information as there are grains of sand on all the world's beaches
Yottabytes (YB)	1,000 Zettabytes	as much information as there are atoms in 7,000 human bodies

https://mynasadata.larc.nasa.gov/basic-page/data-volume-units

Modern Data Science Skills

- Programming in Python
- Statistics
- **Machine Learning**
- Data Analysis



https://ryanswanstrom.com/2016/11/28/data-scientists-data-engineers-software-engineers-the-difference-according-to-linkedin/

Early History of Artificial Intelligence

- **Early Foundations (1940s - 1950s):**
  - Warren McCulloch & Walter Pitts: Developed the first neural network model in 1943.
  - Alan Turing (1950): Introduced the Turing Test to evaluate machine intelligence.
- **The Birth of AI (1956):**
  - John McCarthy: Coined "Artificial Intelligence" at the Dartmouth Conference, initiating AI as a field.
- **The Golden Years (1950s - 1970s):**
  - Arthur Samuel: Created the first self-learning program (checkers) in 1959.
  - Joseph Weizenbaum: Developed ELIZA, an early NLP program, in 1966.
- **The AI Winter (1970s - 1980s):**
  - Decline in AI research due to funding cuts and unmet expectations.

Modern AI: Resurgence, Deep Learning, and Generative AI

- **The Resurgence (1980s - 2000s):**
  - *Backpropagation* (1980s): Revitalized neural networks, with breakthroughs in machine learning.
  - *IBM's Deep Blue* (1997): Defeated chess champion Garry Kasparov, and demonstrated AI's capabilities.
- **The Deep Learning Revolution (2010s):**
  - AlexNet (2012): Pioneered *deep learning for image recognition*, winning the ImageNet competition.
  - AlphaGo (2016): Developed by Google DeepMind, defeated Go champion Lee Sedol, and showed AI's strength in complex strategic games.

Modern AI: Resurgence, Deep Learning, and Generative AI

- **Generative AI and GPT (2020s):**
- Generative Adversarial Networks (GANs):
  - Creation of realistic images, music, and more through adversarial training.
- GPT Models (2018 - Present):
  - Developed by OpenAI, GPT-4 and GPT-5 are advanced language models capable of generating human-like text.
- **The Future of AI:**
- Ethical AI, explainability, and broader applications (e.g. quantum computing and personalized medicine).
- Foundational Models

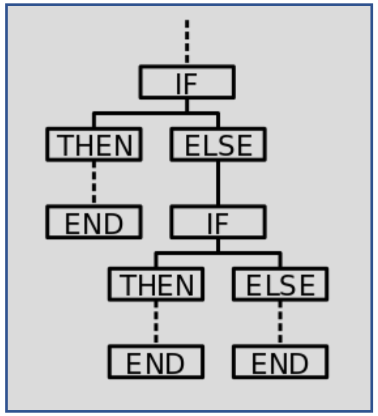
What is Machine Learning?

Machine learning is the subfield of computer science that gives "**computers the ability to learn without being explicitly programmed**."

Arthur Samuel
American computer scientist, coined the term "machine learning" at IBM in 1959

How Does Machine Learning Work?

Feature extraction



A set of rules

Dog

Apple

Flower

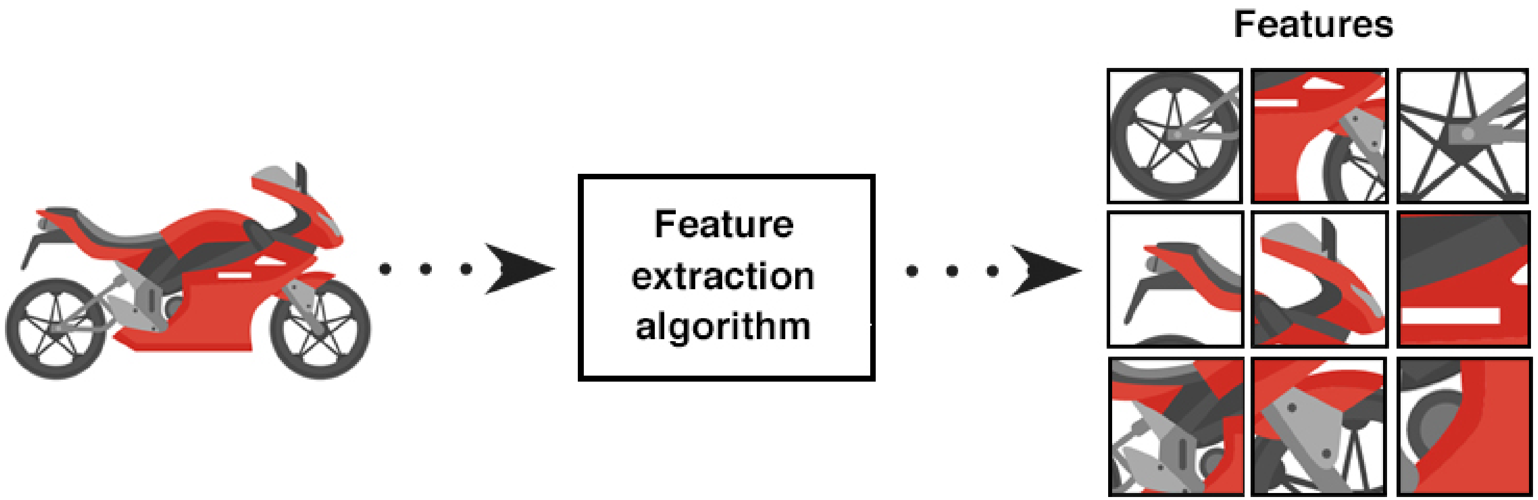
Book

Tree

Conventional Methods

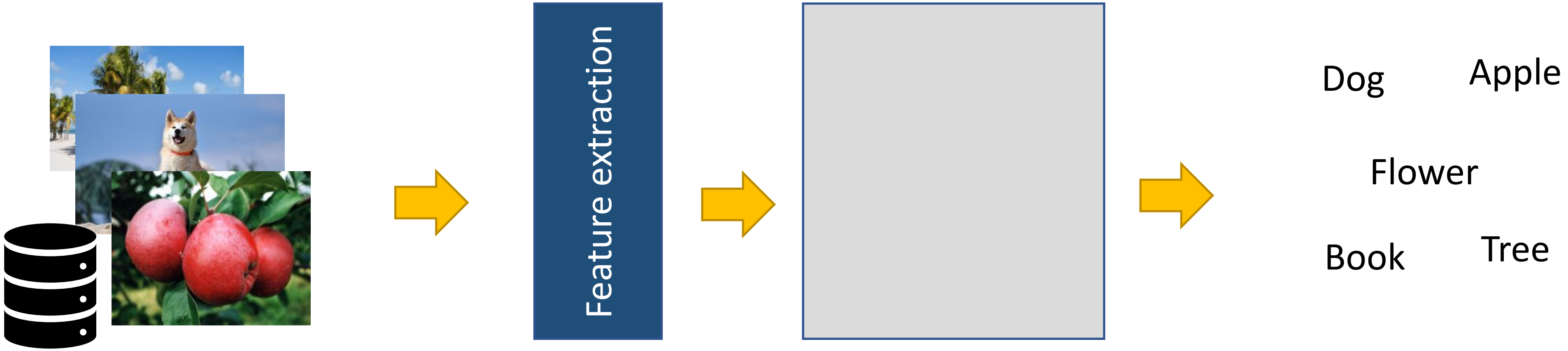
What Is a Feature?

- In computer vision, a feature is a measurable piece of data in your image which is unique to this specific object.
- Example: distinct color in an image, a specific shape such as a line, edge, or an image segment.



https://freecontent.manning.com/the-computer-vision-pipeline-part-4-feature-extraction/

The Way Machine Learning Works

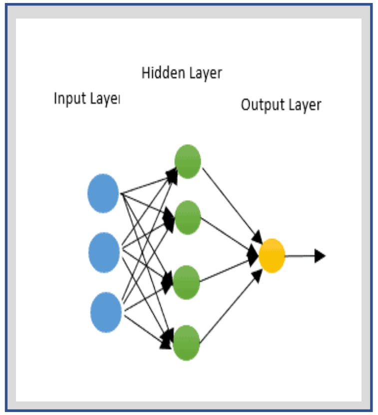


Key Ideas:

- Machine Learning algorithms automatically identify patterns in data.
- These patterns are then used to make predictions or decision.

The Way Machine Learning Works

Feature extraction



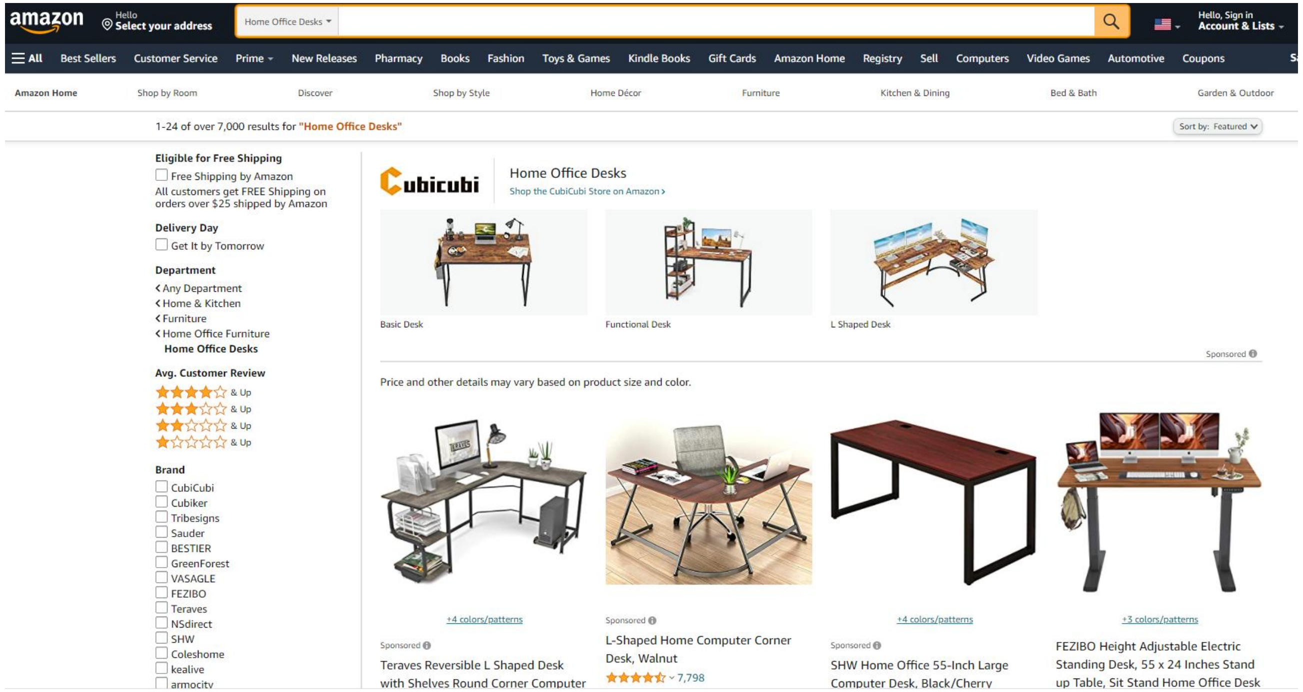
A machine Learning model

Dog Apple

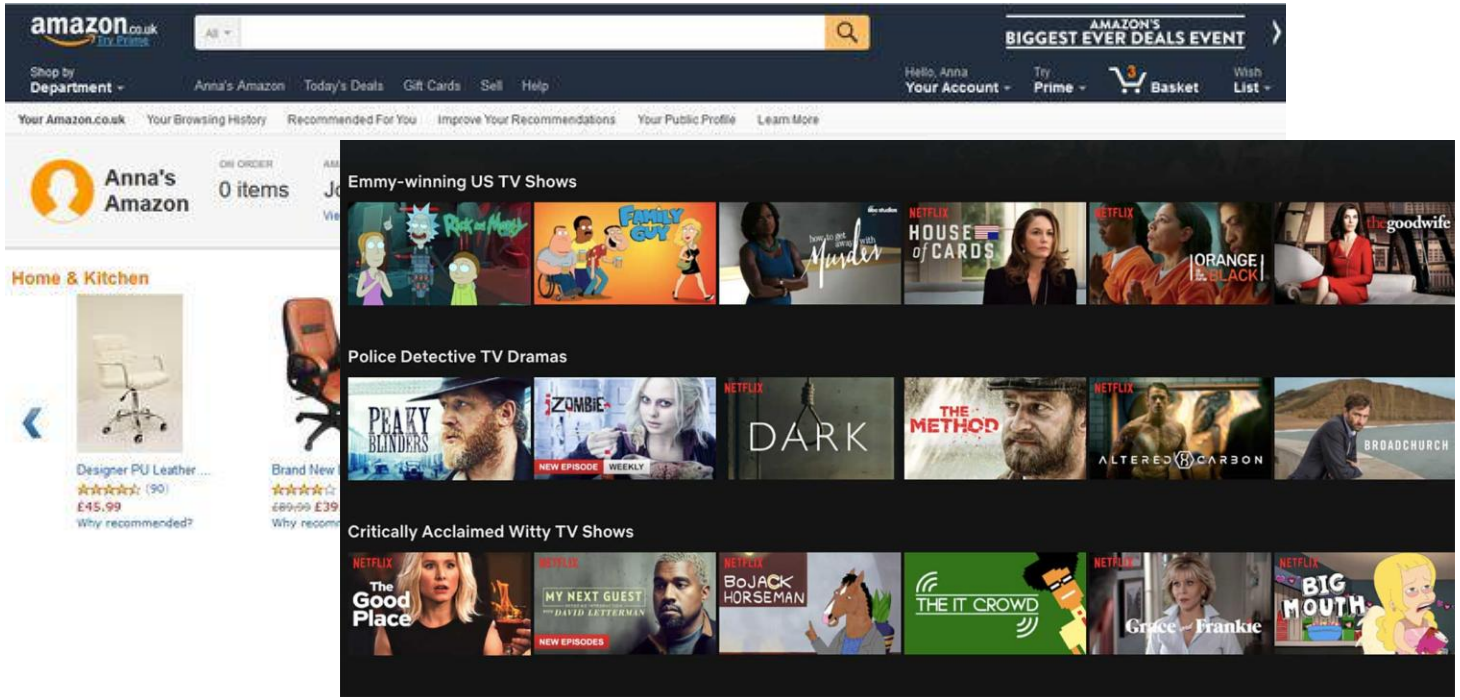
Flower

Book Tree

Examples of Machine Learning in Daily Life



Examples of Machine Learning



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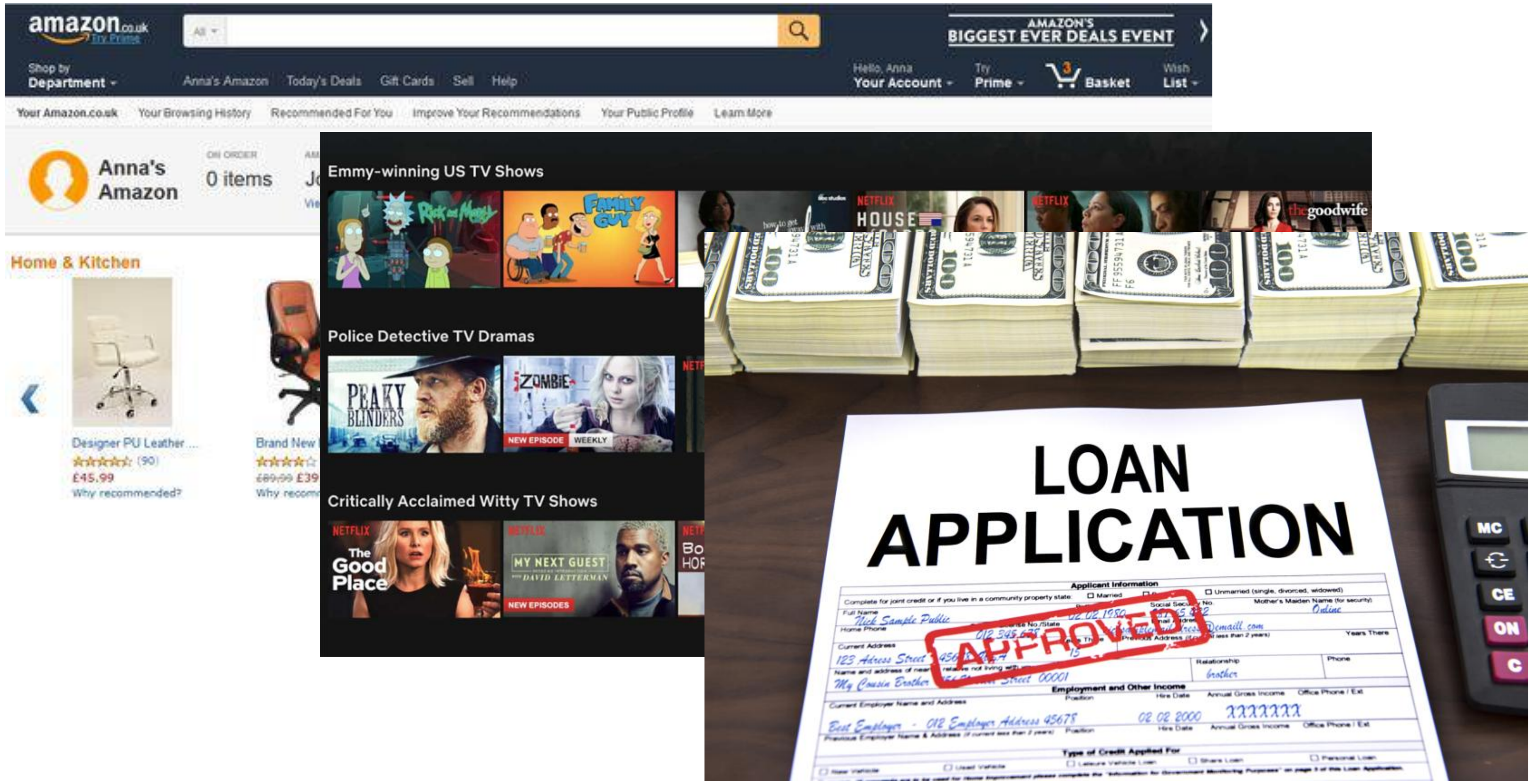

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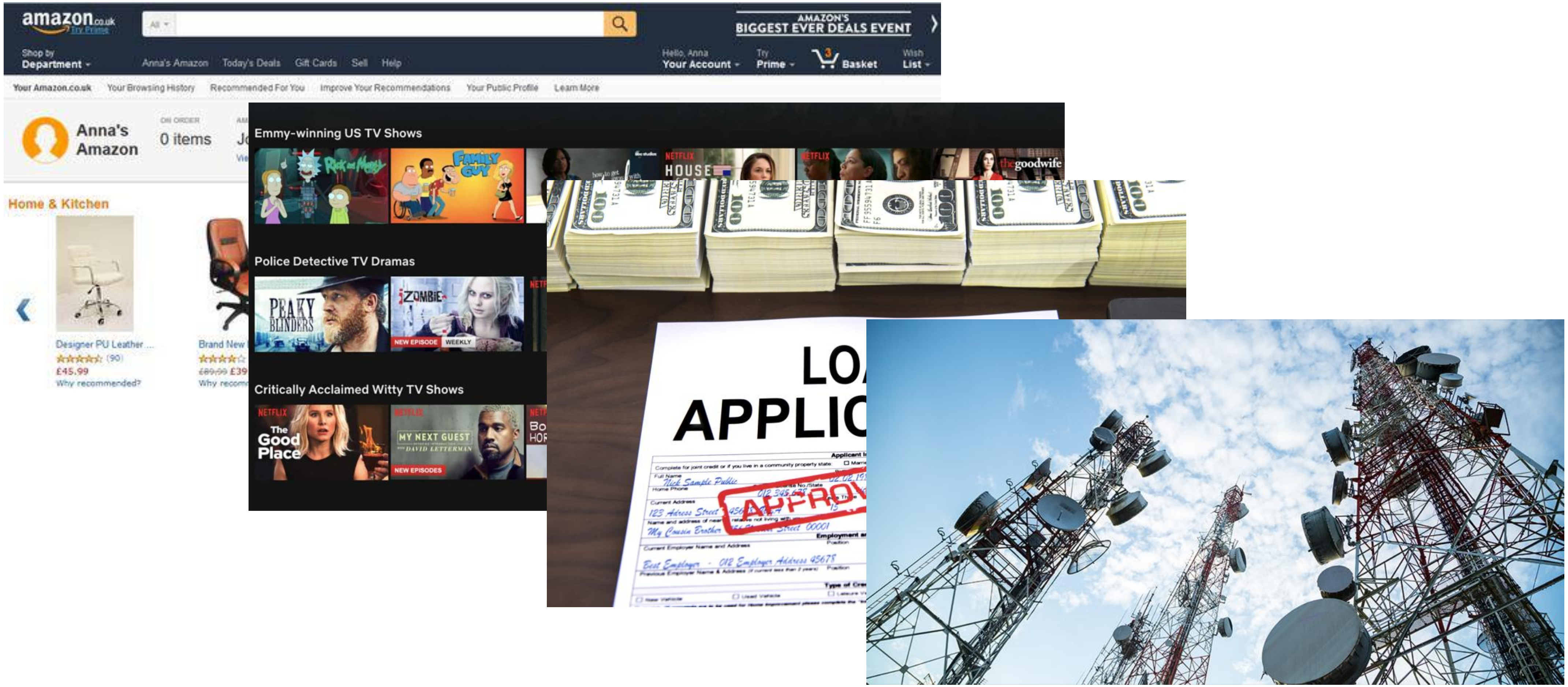

Scope of Topics

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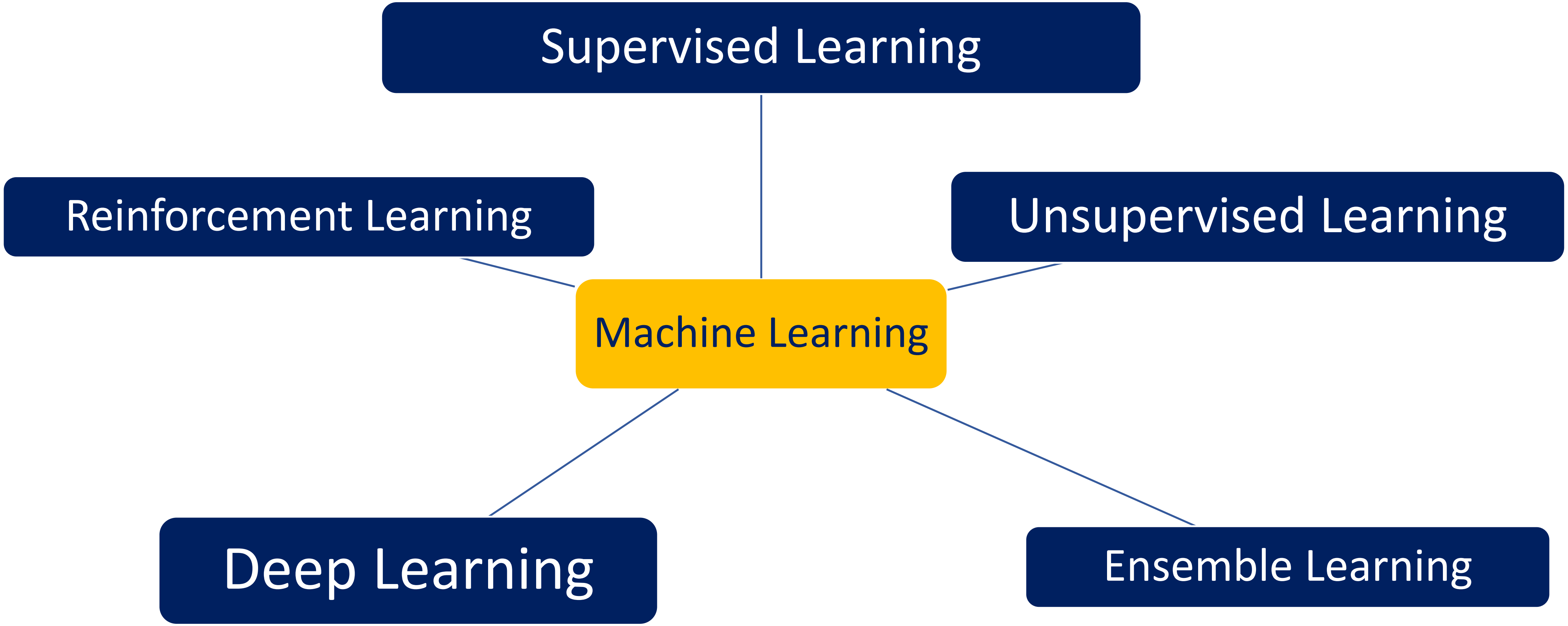
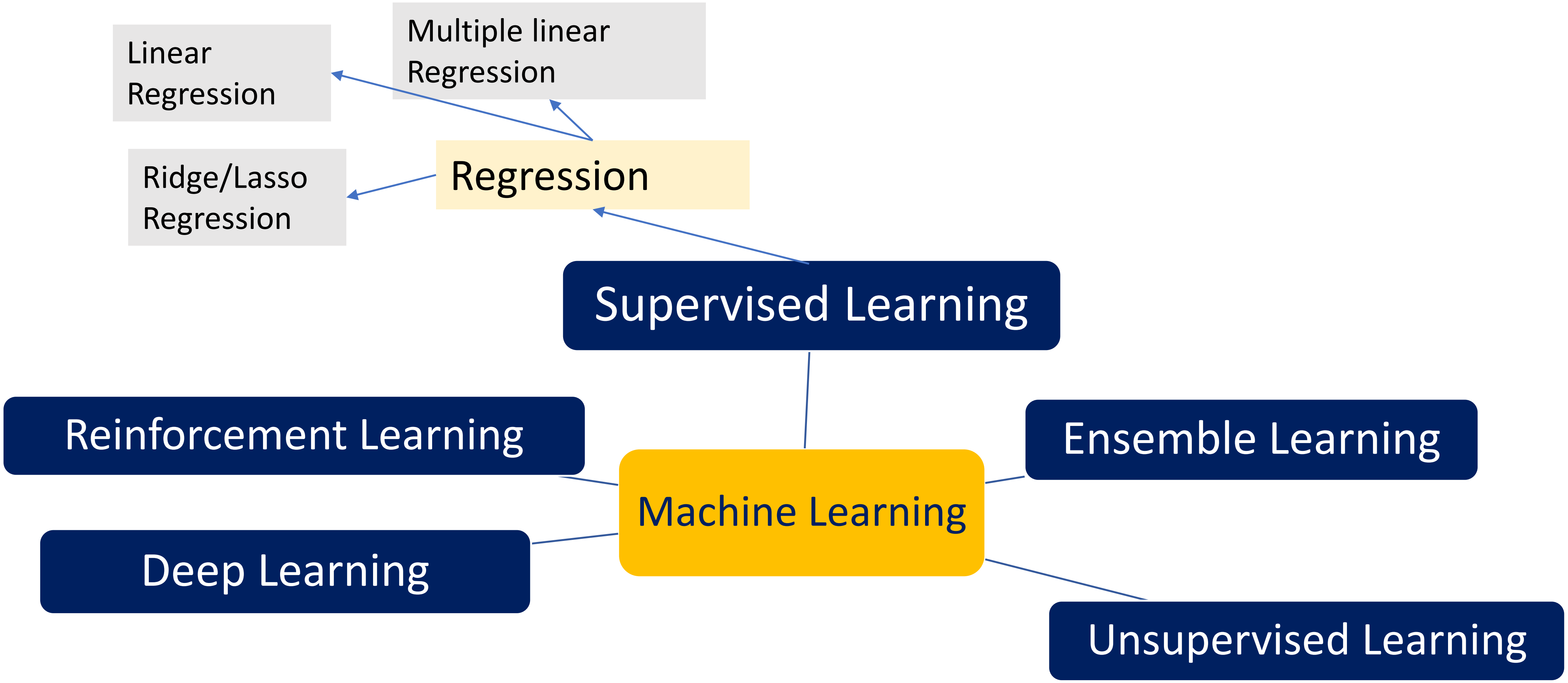
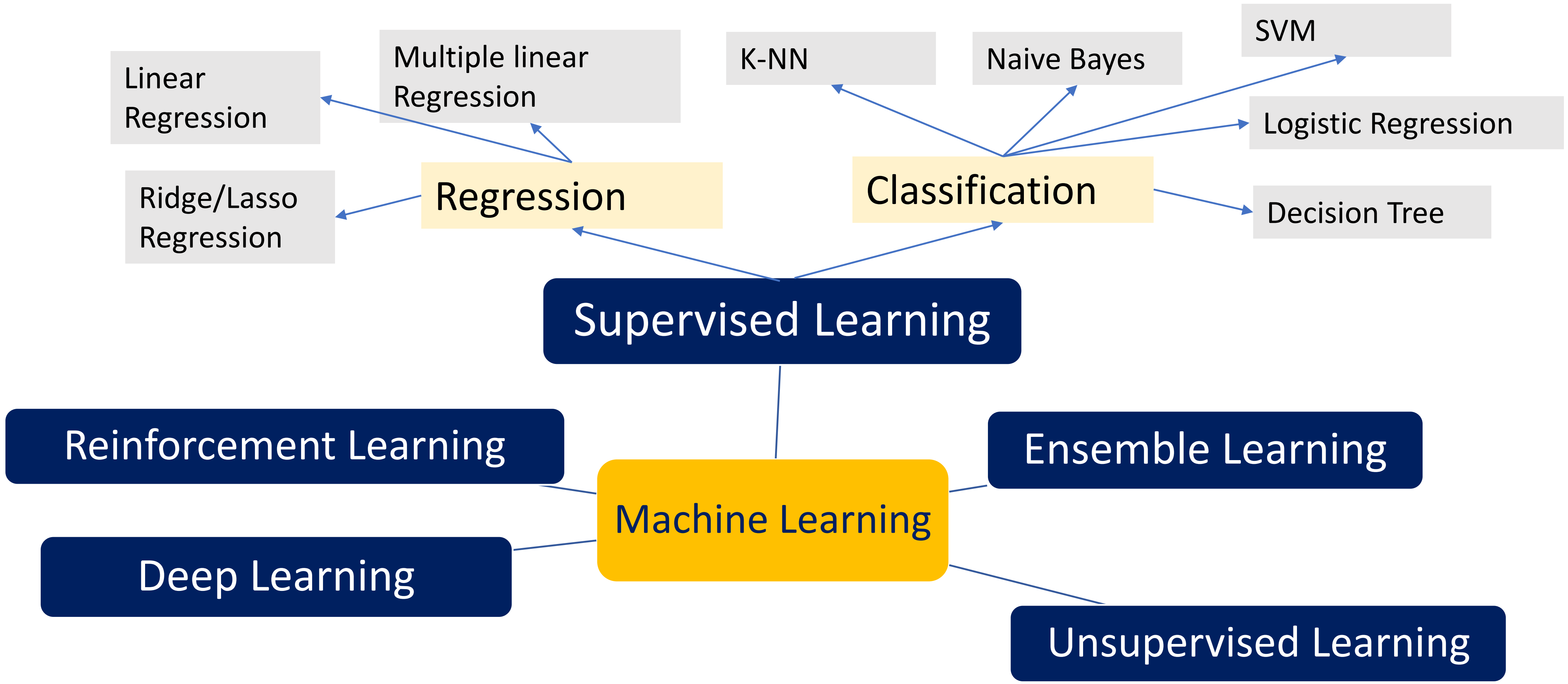



Linear Regression
Multiple linear Regression
Ridge/Lasso Regression
Regression
Supervised Learning
Reinforcement Learning
Ensemble Learning
Machine Learning
Deep Learning
Unsupervised Learning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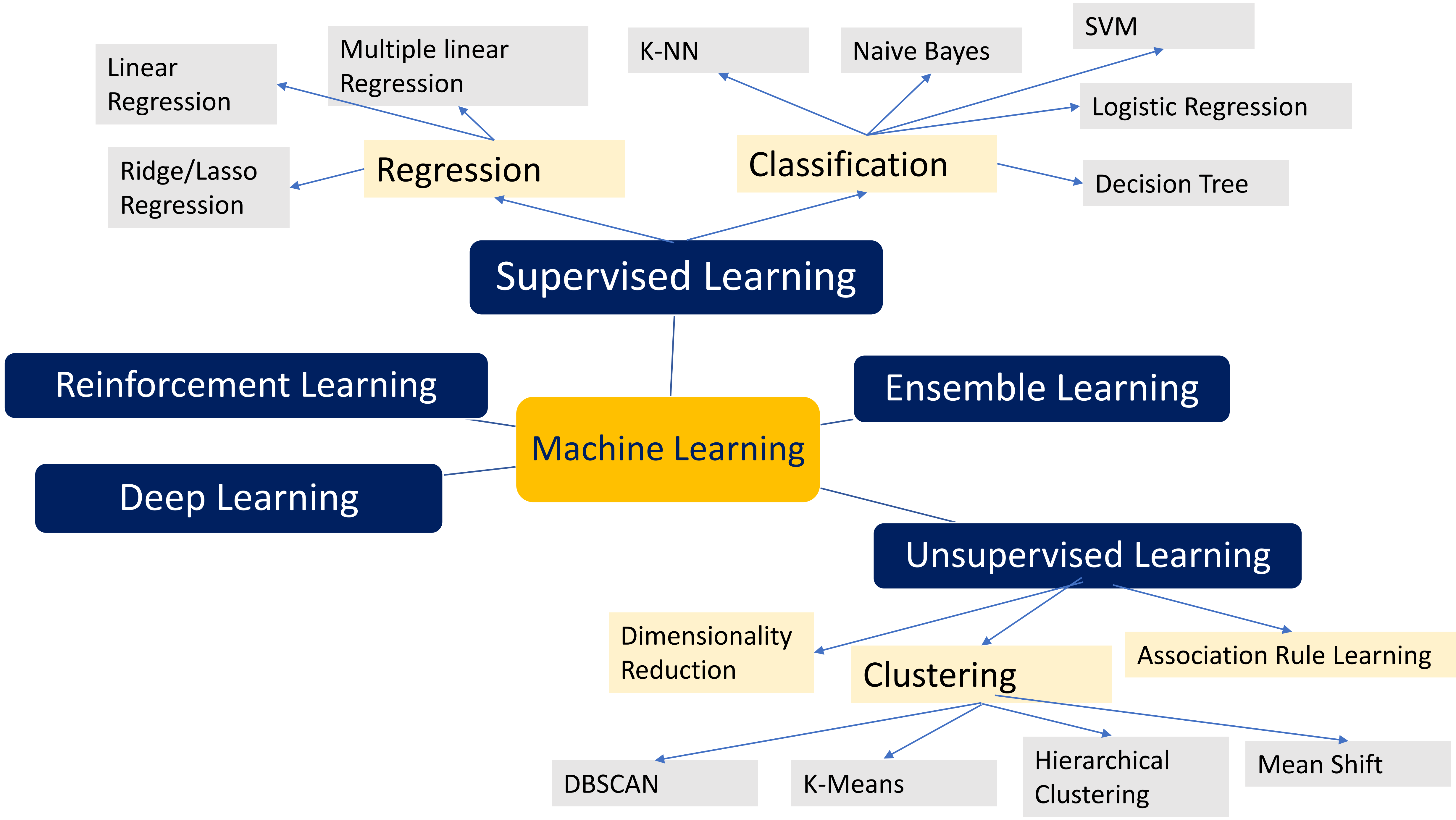

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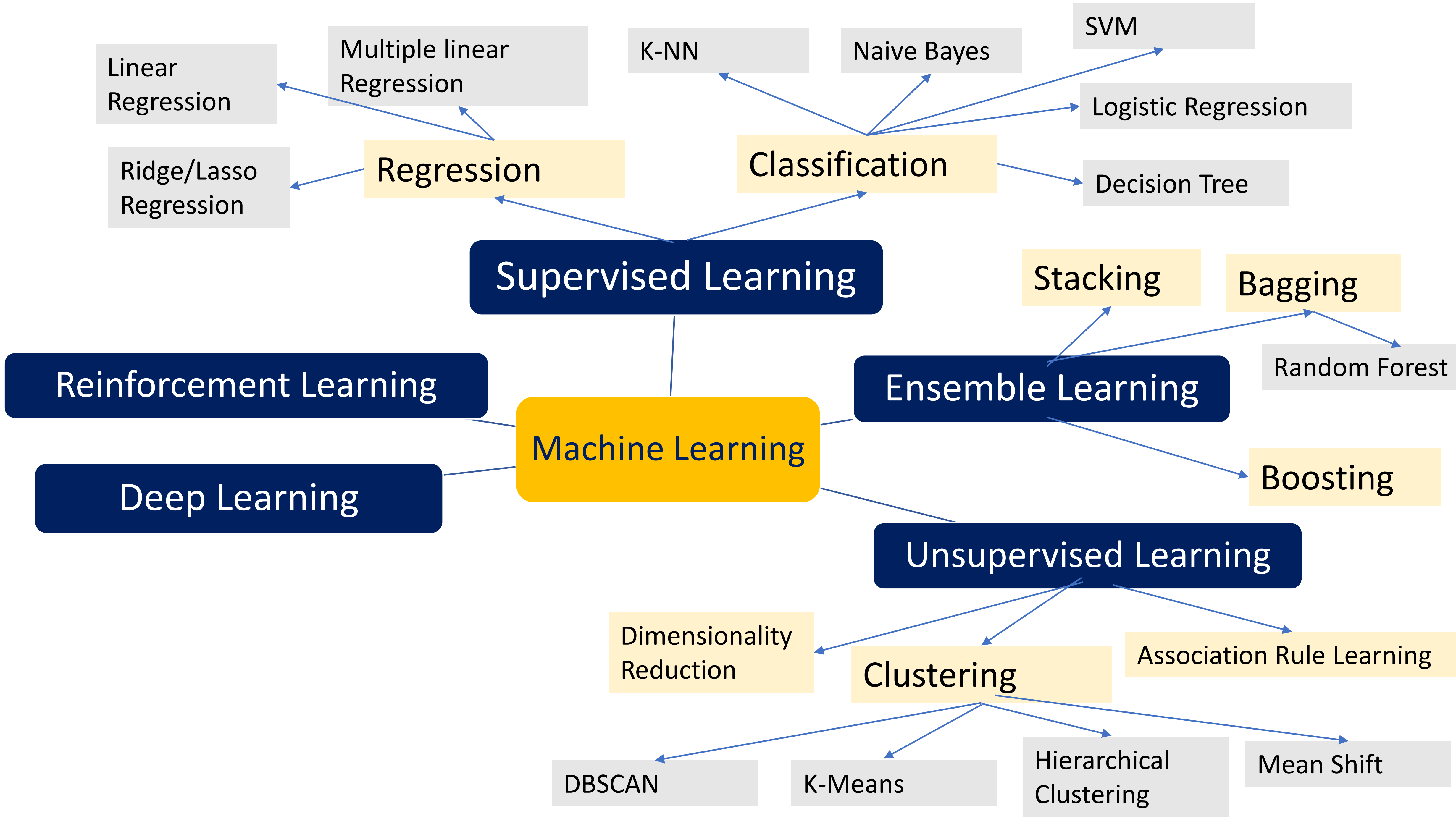
Deep Learning
Unsupervised Learning
Dimensionality Reduction
Clustering
Association Rule Learning
DBSCAN
K-Means
Hierarchical Clustering
Mean Shift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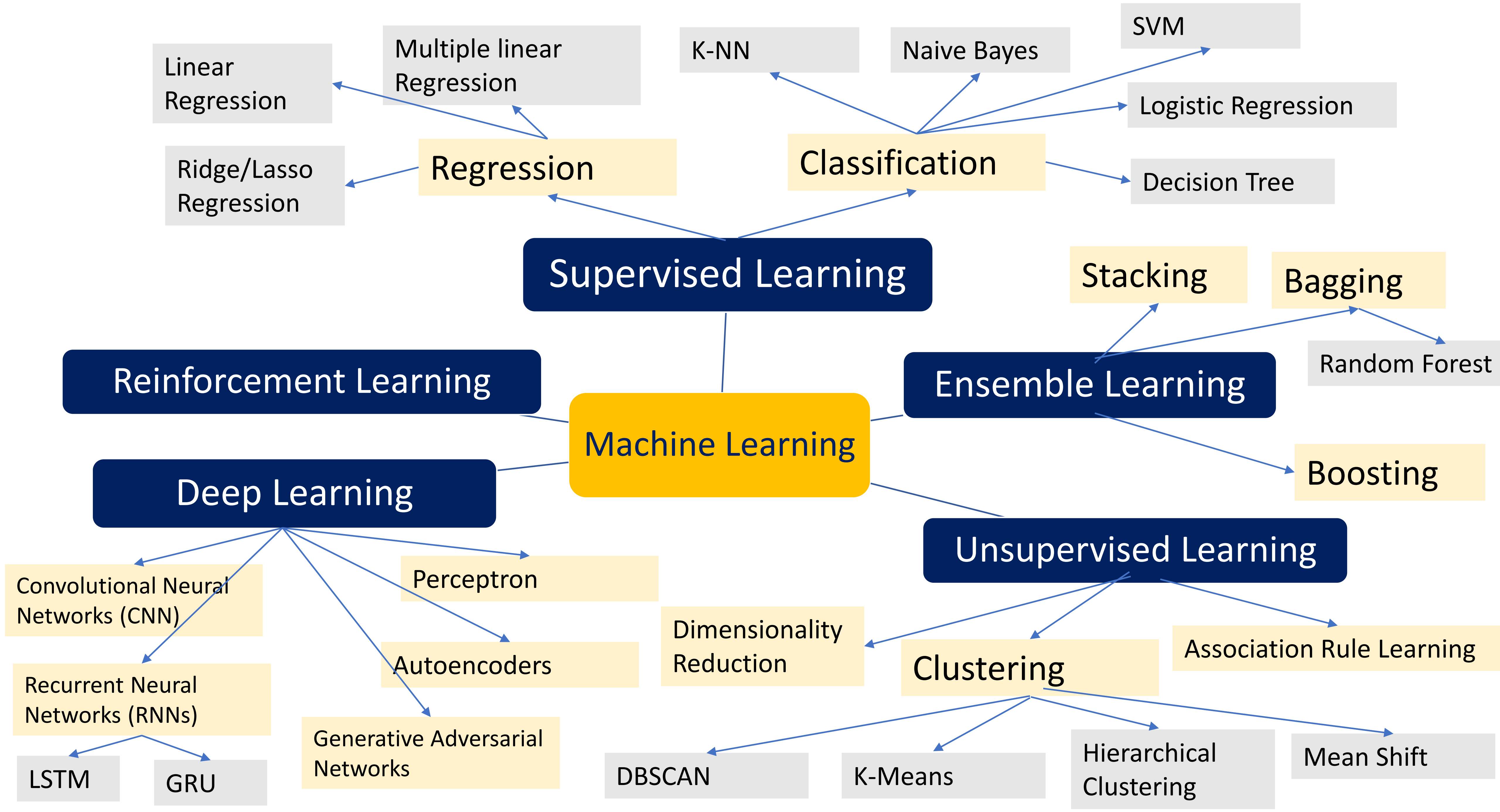
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Random Forest
Boosting
Deep Learning
Convolutional Neural Networks (CNN)
Recurrent Neural Networks (RNNs)
LSTM
GRU
Perceptron
Autoencoders
Generative Adversarial Networks
Unsupervised Learning
Dimensionality Reduction
Clustering
Association Rule Learning
DBSCAN
K-Means
Hierarchical Clustering
Mean Shift

Major Machine Learning Techniques

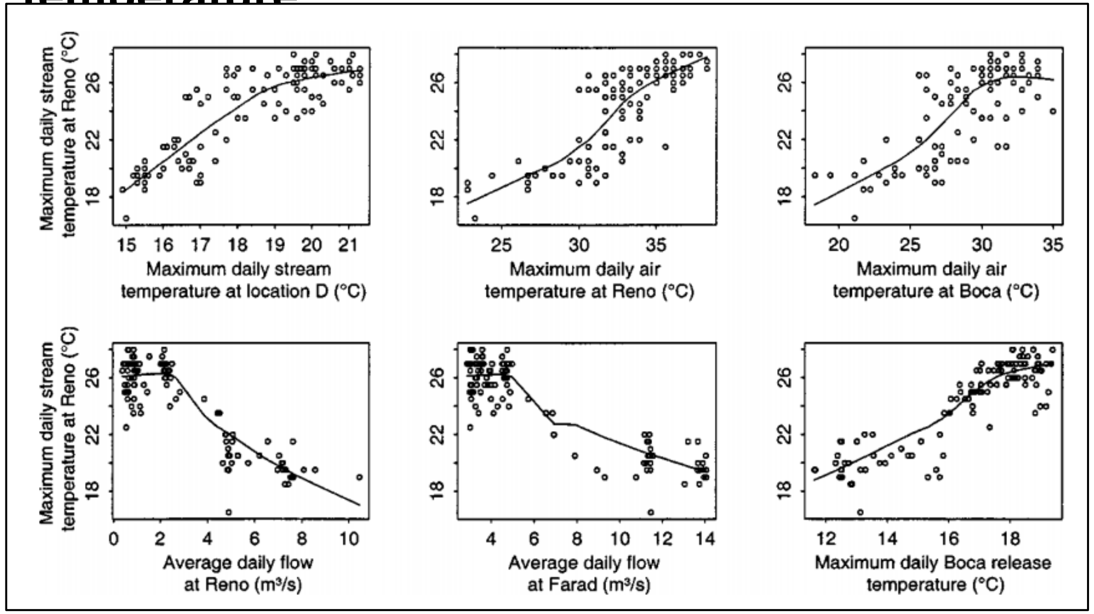
- Regression/Estimation
- Classification
- Clustering
- Associations

Major Machine Learning Techniques

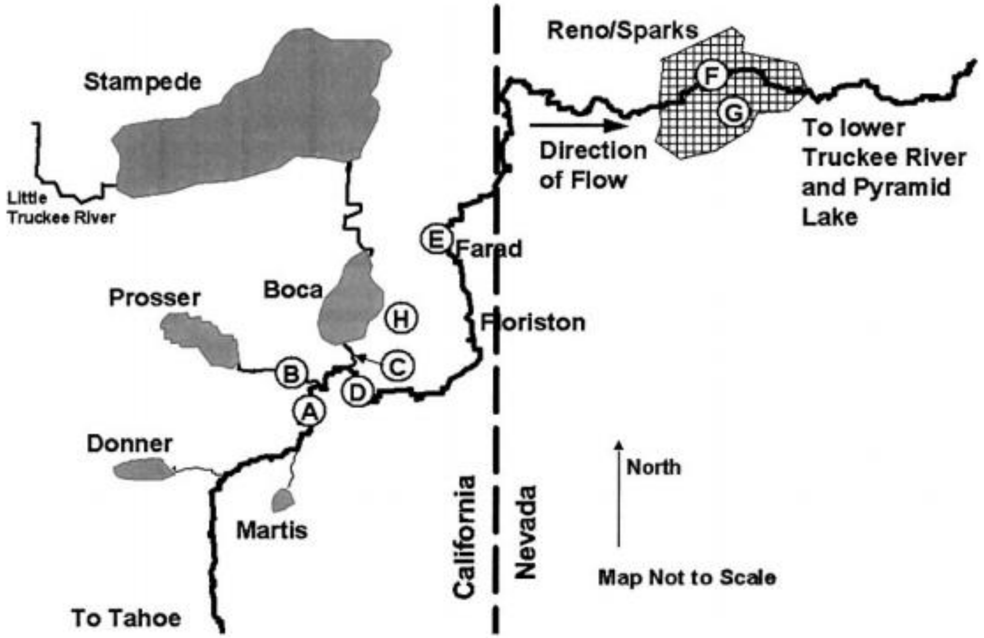
- Regression/Estimation
  - Predicting continuous values
- Classification
- Clustering
- Associations

Example: Regression

Example: Predicting Truckee River **stream temperatures** at Reno by using a simple linear regression relationship based on **flow** and **air temperature**



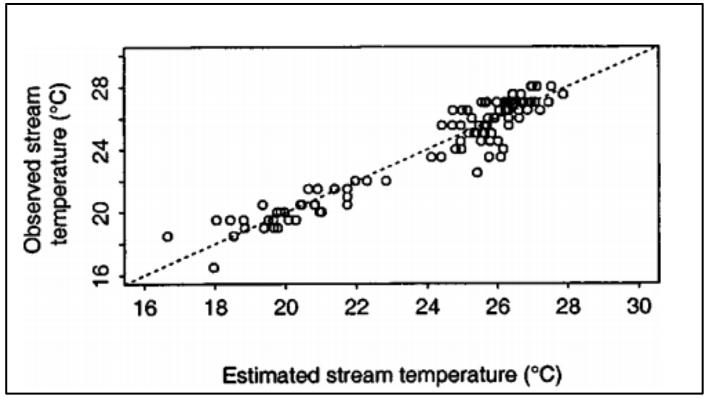
Data used as predictors



A linear regression using the predictors selected has the following equation:

$$\hat{T} = a_0 + a_1 T_{\mathrm{Air}} + a_2 Q$$

where T_{Air} = air temperature at Reno; and Q = flow at Farad.



Neumann, D. W., Rajagopalan, B., & Zagona, E. A. (2003). Regression model for daily maximum stream temperature. *Journal of Environmental Engineering*, *129*(7), 667-674.

Major Machine Learning Techniques

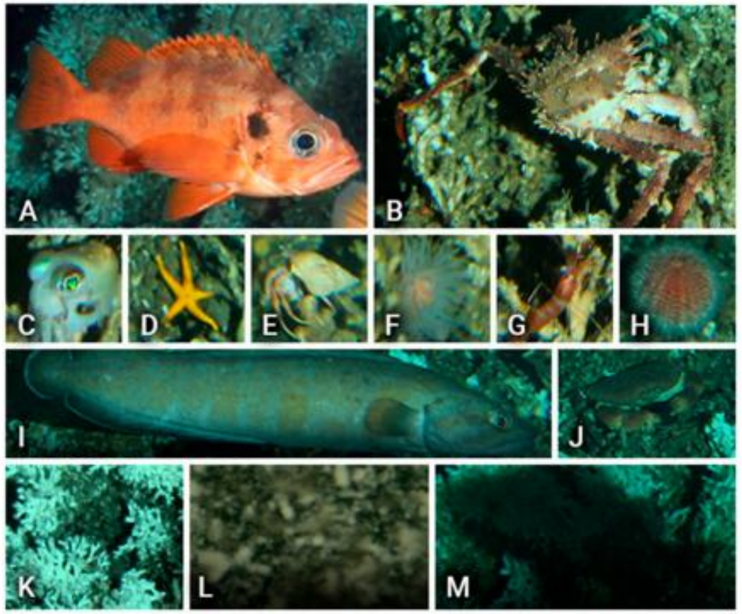
- Regression/Estimation
  - Predicting continuous values
- Classification
  - Predicting the item class/category of a case
- Clustering
- Associations

Example: Classification

- Underwater Animal Detection and Classification

An example of video-detected species used for building the training dataset for reference at automated classification.

Class (alias)	Species Name	# Specimens per Species in Dataset	Image in Figure 2
Rockfish	*Sebastes* sp.	205	(A)
King crab	*Lithodes maja*	170	(B)
Squid	*Sepiolidae*	96	(C)
Starfish	*Unidentified*	169	(D)
Hermit crab	*Unidentified*	184	(E)
Anemone	*Bolocera tuediae*	98	(F)
Shrimp	*Pandalus* sp.	154	(G)
Sea urchin	*Echinus esculentus*	138	(H)
Eel like fish	*Brosme brosme*	199	(I)
Crab	*Cancer pagurus*	102	(J)
Coral	*Desmophyllum pertusum*	142	(K)
Turbidity	-	176	(L)
Shadow	-	101	(M)



An example of video-detected species used for building the training dataset for reference at automated classification: (A) Rockfish (Sebastes sp.), (B) king crab (Lithodes maja), (C) squid (Sepiolidae), (D) starfish, (E) hermit crab, (F) anemone (Bolocera tuediae), (G) shrimp (Pandalus sp.), (H) sea urchin (Echinus esculentus), (I) eel-like fish (Brosme brosme), (J) crab (Cancer pagurus), (K) coral (Desmophyllum pertusum), and finally (L) turbidity, and (M) shadow.

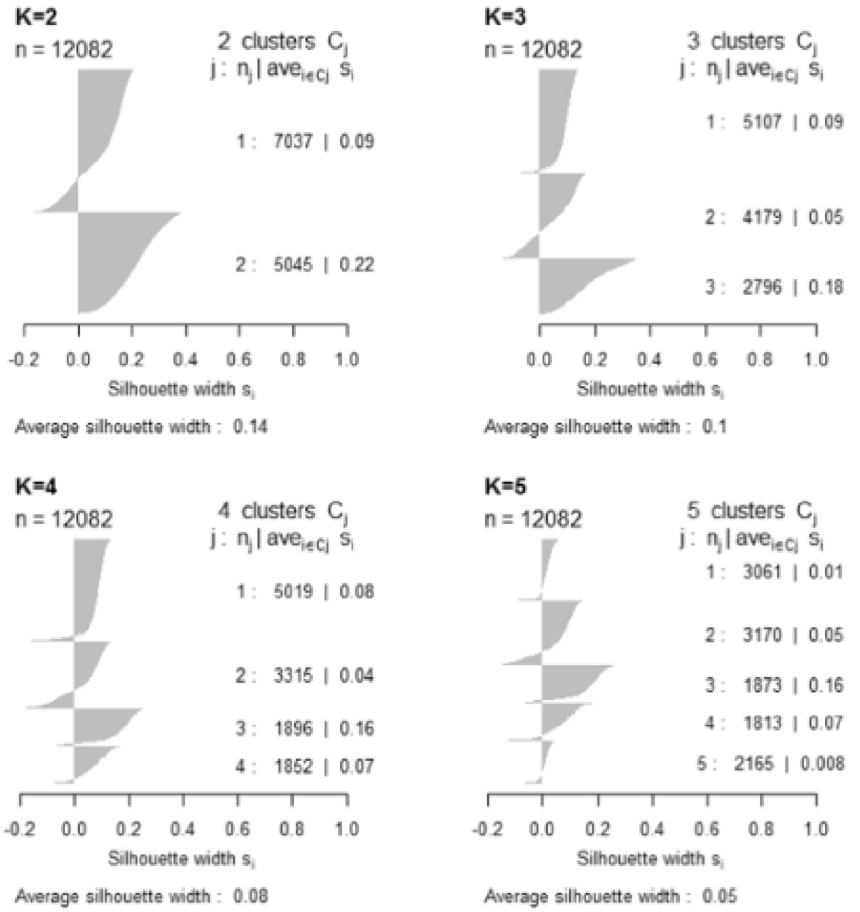
Lopez-Vazquez, Vanesa, et al. "Video image enhancement and machine learning pipeline for underwater animal detection and classification at cabled observatories." *Sensors* 20.3 (2020): 726.

Major Machine Learning Techniques

- Regression/Estimation
  - Predicting continuous values
- Classification
  - Predicting the item class/category of a case
- Clustering
  - Finding the structure of data, summarization
- Associations

Examples: Clustering

- Cluster households based on their energy consumption.
- **Dataset:** The 2009 Residential Energy Consumption Survey (RECS) data, collected by the US Energy Information Administration (EIA)
- How many clusters?



Cluster analysis results on the total electricity consumption data. Silhouette analysis suggests 2 clusters.

Mashhadi, Ardeshir Raihanian, and Sara Behdad. "Discriminant effects of consumer electronics use-phase attributes on household energy prediction." *Energy policy* 118 (2018): 346-355.